

DARREN GAO

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EDUCATION

University of Maryland, College Park

Expected May 2030

Advanced Cybersecurity Experience for Students (ACES) Honors College

Major: Computer Science

Related Coursework: HACS208A, HACS100, CMSC 131, MATH 141

Eleanor Roosevelt High School — Science and Technology Program

Greenbelt, MD

Weighted GPA: 4.417 | Unweighted GPA: 3.964

EXPERIENCE

Research Intern, Institute for Systems Research

June 2025 – June 2026

University of Maryland

College Park

- Conduct research under Dr. Huan Xu on the intersection of artificial intelligence and Augmentative and Alternative Communication (AAC), co-developing findings for publication
- Design and build a bilingual AAC prototype intended to expand communication access for individuals with speech impairments, implementing machine learning algorithms to enhance predictive text and user interaction
- Engineer a personalization engine combining frequency analysis, recency weighting, and Markov-chain-based context modeling to rank phrase suggestions in real time across English and Spanish
- Integrate speech recognition and text-to-speech capabilities using multithreaded processing to enable hands-free, bilingual voice interaction without compromising GUI responsiveness

Projects

Smart Money Tracker

- Developing an automated stock-monitoring application that tracks SEC filings from 100 prominent investors and institutions, detecting new positions, portfolio changes, and insider transactions.
- Building a real-time filing analysis and email notification system using Python, FastAPI, and PostgreSQL to identify significant investment activity and deliver actionable alerts.

Blue Flappy Bird

- Engineered the complete game architecture for "Blue Flappy Bird," implementing custom gravity simulation, hit-box collision detection, and a dynamic score-tracking system
- Designed and integrated all user interface elements and interactive mechanics to create a seamless player experience

ACTIVITIES

ERHS Japanese National Honor Society

June 2024 – May 2026

Vice President, June 2025 – May 2026

- Served as the President's primary support and a core member of the executive cabinet, sustaining chapter operations
- Coordinated and delegated responsibilities across chapter members, cabinet officers, the faculty sponsor, and school staff

Academic Advisor, June 2024 – May 2025

App Dev Club | Web Dev Bootcamp

September 2026 – Present

- Selected from a highly competitive applicant pool (less than 10% acceptance rate) to participate in an intensive web development bootcamp, building responsive web applications using modern frameworks
- Collaborating with a team of student developers to design, build, and deploy a full-stack web project, utilizing Git/GitHub for version control and agile development workflows.

AWARDS

- First Place, PGCPs 2026 County Math Competition
- Second Place, PGCPs 2025 County Math Competition

TECHNICAL SKILLS

Python, Java, Excel, Git, FastAPI